

AVULA PUNITH KUMAR REDDY

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Professional Summary

Trainee Engineer with nearly 2 years of hands-on experience at Bharat Electronics Limited in the engineering, testing, integration, commissioning, and maintenance of production-grade liquid cooling systems for mission-critical applications. Experienced with PLC-operated demineralized-water cooling units designed to maintain tight temperature tolerances and ensure uniform fluid flow across multi-channel distribution networks. Experienced across both standalone vapor-compression refrigeration circuits (R134a) and centralized chilled-water systems utilizing plate heat exchangers. Responsible for vendor technical discussions, drawing reviews, inspections, performance testing, installation supervision, shipboard integration, commissioning, troubleshooting, maintenance, and system modifications. Proficient in SolidWorks, AutoCAD, GD&T, and DFM, with practical exposure to thermal management, automated cooling systems, equipment reliability, and critical infrastructure.

Technical Skills

- **Cooling & Thermal Systems:** Chilled Water Systems
- **System Integration & Commissioning:** Equipment Installation, Shipboard Integration, Functional Testing, Performance Testing, Commissioning, Troubleshooting, Preventive Maintenance
- **Automation & Controls:** PLC-Operated Cooling Systems, Automatic Temperature Control, Flow Control Systems, System Monitoring & Interlocks
- **Mechanical Design:** 3D Modeling, Assembly Design, GD&T, DFM, Sheet Metal Design, Welding, Engineering Drawing Review
- **CAD/Modeling:** SolidWorks, AutoCAD, CATIA V5
- **Analysis & Simulation:** SolidWorks Simulation, ANSYS (Basic)
- **Engineering & Vendor Management:** Vendor Technical Discussions, Vendor Drawing Review, Vendor Inspection, Technical Coordination, System Modifications & Improvements
- **Tools:** MS Office Suite, SAP ERP

Experience

Bharat Electronics Limited

Hyderabad, Telangana

Trainee Engineer – Mechanical Systems

Feb 2025 – Present

- Managed production-grade, mission-critical shipboard cooling systems circulating demineralized (DM) water to cool high-power defense electronic payloads under stringent operating conditions.
- Operated and evaluated dual cooling architectures: a standalone vapor-compression refrigeration system (R134a) maintaining strict coolant temperature limits, and a centralized chilled-water circuit coupled with plate heat exchangers (PHE) to eliminate separate compressor loops.
- Executed precision hydronic balancing across multi-channel cooling networks using supply headers and calibrated needle valves, ensuring uniform flow distribution and optimal pressure balance.
- Monitored and maintained PLC-automated closed-loop control systems integrating plate heat exchangers, solenoid valves, flow control valves, and circulation pumps for automated thermal stability.
- Led vendor technical discussions, reviewed engineering fabrication drawings, conducted factory acceptance inspections, and contributed to component selection and cooling system modifications.
- Supervised equipment installation, performed functional/performance testing, and executed shipboard integration and commissioning in coordination with naval and shipyard personnel.
- Conducted rapid root-cause troubleshooting, preventive maintenance, and mechanical drawing updates in SolidWorks and AutoCAD adhering to GD&T and DFM standards.

Education

JNTUH UNIVERSITY COLLEGE OF ENGINEERING SULTANPUR

Sangareddy, Telangana

Bachelor of Technology in Mechanical Engineering; CGPA: 6.9/10.0

2024

Sri Chaitanya Jr kalashala

Hyderabad, Telangana

Intermediate; Percentage: 91.4%

2018 - 2020

Unique High School

Hyderabad, Telangana

10th Class; CGPA: 9.7/10.0

2018

Projects

Bladeless Fan using Ion Wind

- Designed, analyzed, and built an experimental bladeless airflow system harnessing corona discharge ionization to produce non-mechanical fluid propulsion.
- Engineered precision 3D CAD assemblies of electrode arrays in SolidWorks, analyzing structural tolerances to maintain uniform electrostatic field distribution.
- Executed systematic experimental trials to map the voltage-to-airflow output curve, optimizing operating electrical parameters for maximum aerodynamic efficiency.
- Fabricated a modular enclosure utilizing FDM 3D printing, integrating high-voltage step-up circuitry with dedicated electrical insulation to eliminate arcing and ensure operational safety.

Automated PET Filament Extruder & Spooling System

- Designed and built an automated recycling system to convert waste PET plastic bottles into continuous, usable 3D printing filament using a custom thermal pultrusion mechanism.
- Engineered a precision bottle-stripping cutter and a motorized, gear-driven spooling mechanism to maintain steady pulling tension and ensure uniform filament winding.
- Implemented closed-loop temperature regulation using an Arduino and thermistor-coupled heating block, maintaining optimal melting temperature to achieve consistent filament diameter and prevent thermal degradation.
- Integrated an Arduino and dedicated motor driver to modulate pulling speed and torque, synchronizing extrusion rate with spool rotation to eliminate filament thinning and breakage.